

Learning to Incentivize: Convergence-Guaranteed Federated Learning via Client Quality Discovery

Jiajun Wang, Jianxiong Guo, *Member, IEEE*, Juncheng Wang, *Member, IEEE*, Xingjian Ding, Deying Li, and Weili Wu, *Fellow, IEEE*

Abstract—Federated learning (FL) is a privacy-preserving distributed machine learning framework where multiple devices collaborate with the assistance of an aggregator. However, the limitations of aggregator communication result in only a portion of clients with high data quality being selected to participate in FL, but the quality of clients' data cannot be evaluated without access to the original data. Most existing methods for selecting clients employ a data quality metric with empirically defined scores, which may select clients with high non-IID degrees, thereby reducing the accuracy and freshness of the model. Furthermore, due to the unknown quality of clients' data, current incentive mechanisms lack FL convergence guarantees, which prevent the client behavior from improving the global model accuracy. To address these issues, in this paper, we propose using the gradient difference as a metric for the quality of clients' data, which can quantify the non-IID degree and contribution potential of each client. We formulate a client selection problem using the Combinatorial Multi-Armed Bandit (CMAB) model and design an effective selection strategy, improving the worst-case regret proof to provide a theoretical guarantee for it. Based on these results, we develop an incentive mechanism by the FL convergence analysis, quantifying the utility functions of the aggregator and clients, and modeling their interaction as a two-stage Stackelberg game. For the non-convex utility function, our method establishes the existence and uniqueness of the Stackelberg equilibrium, thereby enabling the determination of the optimal strategy for maximizing the utility of all participants. Finally, extensive simulation experiments on real-world datasets demonstrate the effectiveness of our proposed method compared to state-of-the-art approaches.

Index Terms—Federated Learning, Client Selection, Data Quality metric, Incentive Mechanism, Convergence, Multi-Armed

This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grant No. 62202016 and No. 62202055, the Guangdong Basic and Applied Basic Research Foundation under Grant No. 2025A1515012843, the Start-up Fund from Beijing Normal University under Grant No. 312200502510, the Internal Fund from Beijing Normal-Hong Kong Baptist University under Grant No. UICR0400003-24 and UICR0200022-25, and the Interdisciplinary Intelligence SuperComputer Center of Beijing Normal University (Zhuhai).

Jiajun Wang and Xingjian Ding are with the College of Computer Science, Beijing University of Technology, Beijing 100124, China. (E-mail: S202375129@emails.bjut.edu.cn; dxj@bjut.edu.cn)

Jianxiong Guo is with the Advanced Institute of Natural Sciences, Beijing Normal University, Zhuhai 519087, China, and also with the Guangdong Key Lab of AI and Multi-Modal Data Processing, Beijing Normal-Hong Kong Baptist University, Zhuhai 519087, China. (E-mail: jianxiong-guo@bnu.edu.cn)

Juncheng Wang is with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. (E-mail: jcwang@comp.hkbu.edu.hk)

Deying Li is with the School of Information, Renmin University of China, Beijing, China. (E-mail: deylingli@ruc.edu.cn)

Weili Wu is with the Department of Computer Science, The University of Texas at Dallas, Richardson 75080, USA. (e-mail: weiliwu@utdallas.edu)

(Jiajun Wang and Jianxiong Guo contributed equally to this work.) (Corresponding author: Xingjian Ding.)

Manuscript received April xxxx; revised August xxxx.

Bandit, Stackelberg Game.

I. INTRODUCTION

WITH the growing concerns about data privacy and security, Federated Learning (FL) has emerged as an effective distributed machine learning framework for modern large-scale machine learning applications [1], [2]. Unlike traditional centralized learning, in FL, the aggregator coordinates multiple clients for multiple global iterations and collaboratively trains the global model without the need for clients to share the original data, thereby mitigating the risk of data leakage [3], [4]. This technique has been widely applied to various fields, e.g., personalized recommendations, healthcare, and autonomous [5], [6], [7]. Through FL, different devices and organizations can collaborate while ensuring data privacy and enhancing model accuracy.

Beyond safeguarding client data privacy, FL offers two additional advantages [8]: it only requires transmitting model parameters, thereby reducing network latency; and it can attract numerous clients to participate, enhancing the global model quality. However, with the growth of the scale of global model parameters, the computational, storage, and communication burdens on the aggregator are increasing. Furthermore, the closer the data distribution in participating clients' datasets aligns with the global data distribution (i.e., the lower its non independently and identically distributed (non-IID) degree [9]), the more it facilitates global model convergence. Consequently, such clients whose data distribution is similar to the global distribution can be considered to have higher data quality. Although increasing the number of clients expands participation, the spatiotemporal and sensor variations among clients cause significant disparities in their non-IID degrees. Enrolling clients without selection, conversely, hinders global model convergence. These two issues diminish the prominence of FL's aforementioned advantages. Therefore, selecting suitable clients for participation becomes crucial. Yet, due to FL's core privacy-preserving feature, the aggregator cannot directly observe the non-IID degree (i.e., data quality) of client datasets. Additionally, without adequate incentives, clients may be reluctant to contribute more data to the FL [10], [11]. To address these issues, incentive mechanisms for FL are typically designed by combining learning techniques (such as Combinatorial Multi-Armed Bandit, CMAB) with game theory approaches (like Stackelberg games).

Existing research often relies on strong assumptions or empirically defined metrics during learning [12], [13], such

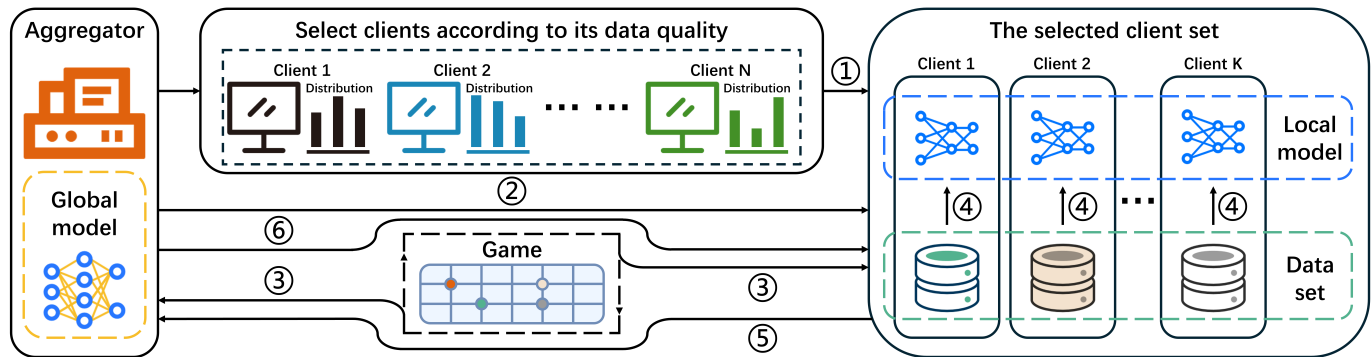


Fig. 1. Our proposed client selection and incentive architecture: In each global iteration, ①: The aggregator selects a subset of clients to participate in FL; ②: The aggregator publishes the global model to the selected clients; ③: The aggregator and the selected client play the game to determine the optimal strategy for each; ④: The selected clients contribute data according to the optimal strategy; ⑤: The selected clients upload their local model; ⑥: The aggregator offers rewards to the selected clients for participating in FL according to the optimal strategy.

as easily observable indicators like latency, computational, or communication resources. However, these approaches lack effective strategies for addressing the core bottleneck: the quality of clients' data, particularly its relation to the degree of non-IID in client datasets. Furthermore, since data quality is unknown, existing incentive mechanisms cannot ensure that the behaviors of aggregators and clients align in a direction conducive to global model convergence [14], [15], [16], [17], [18], [19]. Consequently, incentive design is frequently disconnected from federated learning convergence theory.

In this paper, we investigate an incentive mechanism for FL targeting clients with highly non-IID datasets. Our approach addresses two key questions: (1) *How to design a metric for data quality that enables effective client selection in each FL global iteration without prior knowledge of their data quality?* (2) *How to theoretically ensure that the behaviors of all parties in the incentive mechanism contribute to the convergence of the global model?* These issues are critical in practical FL applications. For instance, edge devices capture highly heterogeneous traffic data due to varying geographic locations, leading to highly non-IID local data. Traditional methods rely on empirically defined quality scores and do not consider model convergence. For metric design, we propose using the upper bound of the gradient difference between a client's local model and the global model as a data quality metric. This metric closely correlates with the discrepancy between a client's data distribution and the global distribution. Based on this data quality metric, we model the client selection problem as a CMAB problem. Then, we design an online learning mechanism for client data quality based on the Combinatorial Upper Confidence Bound (CUCB) algorithm, balancing exploration and exploitation of data quality. For the incentive mechanism, we integrate the Stackelberg game with FL convergence analysis. Positioning the aggregator as the leader and clients as followers, this framework enables the aggregator to reduce rewards offered to clients while maximizing global model accuracy. Our proposed client selection and incentive architecture is illustrated in Fig. 1.

Implementing the proposed incentive mechanism faces three major challenges: First, classic CMAB problems typically

assume the learned metrics lie within the $[0, 1]$ interval, an assumption not met by our designed client data quality metric. This invalidates classical regret bounds. Thus, guaranteeing the regret bound for our CUCB-based approach while extending the metric's range presents a significant challenge. Second, we must integrate federated learning convergence theory into incentive mechanism design to ensure global model convergence. However, formulating utility functions that guarantee convergence poses a distinct challenge. The global model's performance depends on multiple factors (e.g., iteration count, client data contributions, and local data quality), making it difficult to establish equilibrium conditions satisfying both the aggregator and participating clients. Third, utility functions incorporating FL convergence theory often exhibit structural complexity, such as containing non-convex convergence terms. This complexity makes proving the existence and uniqueness of a Stackelberg Equilibrium within the incentive mechanism inherently difficult. Therefore, efficiently identifying this equilibrium and deriving the optimal strategies for all parties constitutes a major challenge.

To address these challenges, we leverage Hoeffding's inequality to establish a sublinear regret bound (scaling with the total number of global iterations T) for selecting K clients per FL iteration. Subsequently, employing backward induction and the Karush-Kuhn-Tucker (KKT) conditions, we analyze the functional properties of the aggregator's utility function. This analysis proves the existence and uniqueness of optimal decisions across different feasible decision sets for the aggregator. Building on this, we demonstrate the existence and uniqueness of the Stackelberg Equilibrium and derive closed-form expressions for the optimal strategies of both the aggregator and the clients. The main contributions are summarized as follows:

- To the best of our knowledge, we pioneer the study of client selection in FL under unknown data quality. We propose using gradient difference as a novel data quality metric and formulate the client selection problem via a CMAB model, offering new perspectives for online learning of data quality and quantifying client contributions. Our CUCB-based framework effectively

TABLE I
WORK COMPARISON ON CLIENT SELECTION AND INCENTIVE MECHANISM

Work	Client selection	Selection criterion	Criterion metric	Incentive	Convergence
[20]	UCB strategy	Data quality	Empirical score	Auction	×
[21]	Thompson sampling	Model similarity	Euclidean distance	×	×
[22]	PROMETHEE[23]	system & network, data quality	Empirical score	×	×
[24]	UCB strategy	Training delay	Time	×	×
[25], [26], [27]	×	×	×	Stackelberg	×
[28], [29], [30]	×	×	×	Auction	×
[31]	×	×	×	Contract	×

balances exploration and exploitation to ensure global model accuracy.

- We design the first Stackelberg utility function that explicitly incorporates the FL convergence upper bound. Complementing this, we construct client utility functions based on their contribution to the global model, thereby designing a coherent incentive mechanism. This effectively links client selection with incentive participation.
- Leveraging backward induction, we derive the KKT conditions to solve the global optimization challenge posed by the complex utility function embedding a non-convex convergence upper bound. This approach allows us to prove the existence and uniqueness of the Stackelberg Equilibrium. Subsequently, we design an efficient algorithm to find near-optimal strategies achieving this equilibrium between the aggregator and clients. Furthermore, we establish a sublinear worst-case regret bound for the CUCB-based client selection framework.
- We conduct extensive experiments on real-world datasets across all non-IID degrees (α) to evaluate our proposed incentive mechanism. Compared to custom baselines tailored to our problem, our proposed mechanism has a faster and more stable convergence rate, and reduces the regret by at least 50.41%.

Organization. We first summarize the works related to this paper in Section II. We then introduce our FL System and formulate the problem in Section III. Next, in Section IV and Section V, the design of our algorithm and the corresponding theoretical analysis are presented in detail, respectively. We carry out the performance evaluation in Section VI and conclude the paper in Section VII.

II. RELATED WORK

Recognizing the importance of selecting the appropriate client to participate in FL for the global model accuracy, much effort has been devoted to designing various client selection frameworks [32], [33]. Guo *et al.* [20] proposed a customer selection algorithm based on CMAB and an incentive mechanism based on auction. The winning bidder was determined according to the UCB quality bid ratio, and key payment was adopted as the payment method. Lu *et al.* [21] quantified the similarity between the local model and the server side based on the fuzzy mathematics strategy, modeled the FL system as an MAB problem, and solved it based on the

Thompson sampling strategy. Tahir *et al.* [22] solved the client selection in FL as a multi-criteria optimization problem. By using the PROMETHEE method, the clients are ranked based on their applicability to the given FL task, while considering various criteria such as system resources, network conditions, and data quality. Beyond client selection, FL architectures like Hierarchical FL [34] and Multi-modal FL [35] are also effective approaches for handling data heterogeneity. In these existing studies, regardless of whether the aggregator is known or not, the criterion for selecting clients mostly fails to reflect the non-IID degree of the clients' dataset. When data quality is used as the selection criterion, its metric is generally set as empirically defined scores rather than the parameters in the FL system that can reflect the data quality of the client. This may not handle clients with high non-IID degree datasets, which can reduce the accuracy and objectivity of the selection.

Enormous theories have been put into practice in the design of FL incentive mechanisms, such as game theory [36], [37], [38], auction theory [39], [40], contract theory [41], cooperative games [42], etc. Li *et al.* [25] constructed the FL incentive mechanism based on the Stackelberg game. Le *et al.* [28] constructed an FL incentive mechanism including a single base station (BS) and multiple mobile users based on auction theory. Chen *et al.* [26] constructed a multi-factor FL incentive mechanism based on the Stackelberg game and designed a Top-K cost selection algorithm based on reverse auction, effectively reducing the screening cost of the client side. He *et al.* [27] classified the clients to form multiple local FL systems based on the similarity of the client's dataset, and the model is aggregated by the cluster heads of each group of clients. Finally, the global model is obtained by the model owner based on the models of each cluster head. Under such an FL architecture, a three-stage Stackelberg game was constructed, and the reinforcement learning algorithm was adopted to solve for the equilibrium point. Xuan *et al.* [31] designed the FL incentive mechanism applicable to multi-task publishers and multi-working nodes based on contract theory. These incentive mechanisms have demonstrated commendable achievements in various aspects. Beyond incentive design, other research has focused on enhancing FL efficiency from different perspectives, such as optimizing resource allocation in IoT networks via federated reinforcement learning [43], and improving over-the-air model aggregation through joint antenna selection and beamforming at the physical layer [44]. However, we observe that existing approaches to designing utility functions for all

parties, particularly those of the aggregator, rarely incorporate convergence analysis for FL, either in the objective functions or the constraints. This gap leaves incentive mechanisms lacking sufficient theoretical guarantees regarding the convergence of the global model. Furthermore, when computing the optimal strategies for participants, some studies employ reinforcement learning techniques, which sometimes fail to guarantee optimal solutions. Finally, the concept of fairness-aware FL, which aims to balance performance with equitable participation, represents another important yet distinct research direction with different objectives.

As mentioned earlier, existing research has made many advancements in the client selection and incentive mechanism design of FL (such as based on game theory, auction theory, etc.). To systematically compare the core methods, key assumptions, and their limitations of these works (especially the combination of assumptions and incentive mechanisms for client-side data quality assessment and convergence theory), we provide a comprehensive comparison of representative literature in Table I. Regarding client selection, mainstream solutions [20], [21], [22] employ empirically defined scores that fail to reflect the degree of non-IID in client datasets when addressing data quality. In incentive mechanism design, existing schemes [25], [26], [28], [31] neglect to incorporate theoretical convergence analysis of FL into their utility function or constraint designs, resulting in a lack of theoretical guarantees for global model performance. These limitations have motivated the research presented in this paper. We propose an Incentive Mechanism based on CMAB and Stackelberg (IMCS), which innovatively introduces gradient differences as a metric for data quality. This data quality metric correlates with EMD, effectively capturing the non-IID degree of client datasets. Pioneeringly integrates the upper bound of the accuracy gap (between the FL global model performance and optimal accuracy) directly into the design of the incentive mechanism's utility function. This constructs a Stackelberg game model with convergence guarantees, providing solid theoretical foundations for participant strategy selection while ensuring efficient convergence of the global model.

III. PRELIMINARIES & PROBLEM FORMULATION

In this section, we first provide an overview of the FL. Then, we introduce the criteria for data quality and the CMAB modeling used for selecting clients with unknown data quality. Finally, we propose an incentive mechanism for FL with unknown data quality clients.

A. Federated Learning

We consider a typical FL framework that contains one aggregator and N rational clients with unknown data quality, denoted by $\mathcal{W} = \{w_1, w_2, \dots, w_N\}$. Each client w_i possesses a local dataset of size D_i . Due to aggregators' limited computational power, storage capacity, and communication bandwidth, it is impossible to involve all clients in FL in each round of global iteration. To reduce communication costs and improve training efficiency, similar to [45], we select K clients to participate in each round of global training. However,

unlike their approach, which selects clients randomly, our method preferentially selects clients with high data quality. We adopt the widely recognized synchronous FL paradigm to enable training on clients' local datasets. This approach involves T iterations of interactions between the aggregator and the clients for optimizing the global model, denoted as $\mathcal{T} = \{1, 2, \dots, T\}$. During each iteration, selected clients perform local model training, and the results of this local training are aggregated to update the global model. In the t -th round, the participating clients are represented by the set $\mathcal{W}_t$, with $|\mathcal{W}_t| = K$. Since clients are unwilling to contribute data to FL without a reward, during the t -th global iteration, the selected client w_i determines the contributed data based on the rewards provided by the aggregator. The number of contributed data samples is denoted as d_i^t . If client w_i participates in FL during the t -th global iteration, then $d_i^t > 0$; otherwise $d_i^t = 0$. This contributed data is randomly sampled from the total dataset of the client w_i .

In a Y class classification problem, each data point (x, y) distributes over $\mathcal{X} \times \mathcal{Y}$ following the distribution $\mathbf{p}$, where $\mathcal{X}$ is a compact space and $\mathcal{Y} = \{1, \dots, Y\}$ is a label space [46]. Let $p_i^t(y = j)$ denote the proportion of data labeled as $y = j$ in the contributed data d_i^t of the client w_i during the t -th global iteration. It is noteworthy that, as the frequency of client w_i 's selection increases, the mean value of $p_i^t(y = j)$ asymptotically approaches the ratio of the data corresponding to label $y = j$ to the total data owned by client w_i . We use a function $f: \mathcal{X} \rightarrow \mathcal{S}$ to map $\mathbf{x}$ to the probability simplex $\mathcal{S} = \{z | \sum_{i=1}^Y z_i = 1, z_i \geq 0, \forall i \in \mathcal{Y}\}$. Then, f_j denotes the probability for $\mathbf{x}$ belonging to the j -th class. We parameterize f on a hypothetical class θ , i.e., the neural network weights, which are used to aggregate the global model. In the t -th global iteration, the aggregator releases the current global model θ^t to each selected client. The loss function for client w_i can be determined as the widely used cross-entropy loss:

$$F_i(\theta^t) = \sum_{j \in \mathcal{Y}} p_i^t(y = j) \mathbb{E}_{\mathbf{x}|y=j} [\ell_j(\mathbf{x}, \theta^t)], \quad (1)$$

where $\ell_j(\mathbf{x}, \theta^t) = -\log(f_j(\mathbf{x}, \theta^t))$.

The goal of the aggregator is to minimize the global loss function, i.e., the weighted average of each client's loss function. Denoting the total data contributed by all clients participating in FL as $D_t = \sum_{w_i \in \mathcal{W}_t} d_i^t$, we calculate the global loss function as:

$$F(\theta^t) = \sum_{w_i \in \mathcal{W}_t} \left(\frac{d_i^t}{D_t} F_i(\theta^t) \right). \quad (2)$$

In t -th global iteration, each client $w_i \in \mathcal{W}_t$ performs H -step local update via mini-batch Stochastic Gradient Descent (SGD) algorithm on its contributed dataset. Let θ_i^{t+1} and $\theta_i^{t,h}$ denote the client w_i 's local model after the t -th global iteration and after h local update within the t -th global iteration, respectively. Accordingly, we have $\theta_i^t = \theta_i^{t,0}$ and $\theta_i^{t+1} = \theta_i^{t,H}$. For client $w_i \in \mathcal{W}_t$, local SGD in h -th local update performs:

$$\theta_i^{t,h+1} = \theta_i^{t,h} - \eta \nabla F_i(\theta_i^{t,h}; b_i^t), \quad (3)$$

where $b_i^t = d_i^t/H$ represents the number of a mini-batch

data samples [47]. Then, we have the global model update formulation:

$$\theta^{t+1} = \sum_{w_i \in \mathcal{W}_t} \frac{d_i^t}{D_t} \theta_i^{t+1}. \quad (4)$$

B. Client Selection via Data Quality

We mentioned in section III-A that the clients' data quality is what we mainly consider when selecting clients to participate in FL. To describe the data quality of each client $w_i \in \mathcal{W}$, we first define the upper bound of the difference between the global loss function and the local loss function in the t -th global iteration, briefly referred to as the gradient difference, denoted as λ_i^t [48]:

$$\|\nabla F(\theta^t) - \nabla F_i(\theta^t)\| \leq \lambda_i^t. \quad (5)$$

The smaller the gradient difference λ_i^t , the better the performance of the local model trained by client w_i in the t -th global iteration. Let $p^t(y = j)$ represent the fraction of data labeled as $y = j$ within the entire dataset contributed by participating clients. A larger value of $p^t(y = j) - p_i^t(y = j)$ indicates a higher non-IID degree in client w_i 's contributed data. Subsequently, λ_i can be estimated using the Earth Mover's Distance (EMD), which quantifies the heterogeneity of data distributions among clients, such as the non-IID degree [46], as follows:

$$\begin{aligned} & \|\nabla F(\theta^t) - \nabla F_i(\theta^t)\| \\ & \leq \sum_{j \in \mathcal{Y}} \|\nabla \mathbb{E}_{\mathbf{x}|y=j} [\ell_j(\mathbf{x}, \theta^t)]\| |p^t(y = j) - p_i^t(y = j)| \\ & \leq \max_{j \in \mathcal{Y}} \|\nabla \mathbb{E}_{\mathbf{x}|y=j} [\ell_j(\mathbf{x}, \theta^t)]\| \sum_{j \in \mathcal{Y}} |p^t(y = j) - p_i^t(y = j)| \\ & \leq g^{max} \cdot EMD_i^t = \lambda_i^t, \end{aligned} \quad (6)$$

where $g^{max} = \max_{j \in \mathcal{Y}} \|\nabla \mathbb{E}_{\mathbf{x}|y=j} [\ell(\mathbf{x}, \theta^t)]\|$ denotes the maximum expected class loss, and $EMD_i^t = \sum_{j \in \mathcal{Y}} |p^t(y = j) - p_i^t(y = j)|$ denote the EMD of client w_i in the t -th global iteration. Ineq. (6) indicates that the smaller the value of λ_i^t , the smaller EMD_i^t , which means the lower the non-IID degree of client w_i 's dataset. Thus, we use λ_i^t to reflect the client w_i 's data quality in the t -th global iteration. Specifically, the smaller the λ_i^t , the higher the data quality on the client w_i .

Since the data contributed by each selected client in every global iteration is randomly sampled from its local dataset, the λ_i^t of each client across different global iterations are independent and exhibit stochasticity. Furthermore, once a client joins the FL system, its dataset remains fixed, regardless of its participation in specific rounds. Consequently, as the number of global iterations increases, the average of λ_i^t for each client converges to a specific, fixed expectation. Accordingly, we treat the value of $\{\lambda_i^t | w_i \in \mathcal{W}_t, t \in \mathcal{T}\}$ as being independently and identically distributed (i.i.d.) according to an unknown distribution P_i , characterized by an unknown expectation λ_i . Note that, there is $\lambda_i^t \in [\lambda_{min}, \lambda_{max}]$ for any client $w_i \in \mathcal{W}$ and $t \in \mathcal{T}$ because the total dataset for each client is determined. However, due to the privacy-preserving nature of FL, the aggregator lacks knowledge of the dataset composition and the specific values of λ_i^t and λ_i for each

client. This makes it challenging for the aggregator to select clients with high data quality to participate in FL.

TABLE II
FIRST 4 LABELS' DATA PROPORTION OF CLIENT 1 ON THE MNIST DATASET UNDER DIFFERENT α

α	$p_1(y = 1)$	$p_1(y = 2)$	$p_1(y = 3)$	$p_1(y = 4)$
0.1	0.7578	0.0815	0.1475	0.0006
0.5	0.0389	0.0885	0.0196	0.0172
1.0	0.0898	0.3028	0.0087	0.2850

Motivation: We verify through a simple FL system that λ_i^t could reflect the data quality of the client $w_i \in \mathcal{W}_t$. Consider an FL system with 50 clients and an aggregator, and use this system to train 50 iterations on the MNIST dataset. To simulate varying degrees of non-IID across clients, we partition the training dataset among the 50 clients using a Dirichlet distribution by $\alpha = 0.1, 0.5$, and 1.0 . Under different α , the datasets of each client are variant. Taking the client w_1 as an example, Table II shows the proportion of data of the first 4 labels to their total data under different α . The amount of data for each client varies under different α values. In each global iteration, all clients participate, and each contributes 50% of its data.

We first compare the relationship between each client's λ_i and its gradient difference when the model accuracy first exceeds 90% under three different α . As shown in Fig. 2, clients with lower λ consistently exhibit smaller gradient differences, which aligns with the Eqn. (6). Besides, we perform a Chi-squared test to examine whether a significant correlation exists between the λ_i^T of each client and the Wasserstein distance between the client's local data distribution and the global data distribution. The resulting p-values are all below 0.5 ($p = 0.0382$ for $\alpha = 0.1$; $p = 0.0345$ for $\alpha = 0.5$; and $p = 0.0356$ for $\alpha = 1.0$), which confirms a statistically significant association between λ_i^t and the data distribution. Next, we compare the global model accuracy and the average lambda $\bar{\lambda}^t = \sum_{w_i \in \mathcal{W}_t} \lambda_i^t / |\mathcal{W}_t|$ across the three α settings. As shown in Fig. 3, when α is 0.1, 0.5, and 1.0, the global model accuracy reaches above 90% in the 8-th, 6-th, and 4-th global iteration, respectively. Models trained on data with lower non-IID degrees achieve faster convergence, indicating superior data quality. Furthermore, lower non-IID degrees correspond to lower $\bar{\lambda}^t$, demonstrating that lower λ_i^t signifies higher data quality. This observation is consistent with our analysis before. Therefore, λ_i^t can effectively reflect client $w_i \in \mathcal{W}$ data quality and demonstrate an influence on accuracy.

Selecting clients with unknown data quality can be framed as an online learning and sequential decision-making problem, for which the CMAB model is particularly well-suited. A typical CMAB model involves a slot machine with multiple arms, where pulling an arm yields a reward that follows an unknown distribution. A decision-maker selects a set of arms in each global iteration, guided by a bandit policy, to maximize cumulative rewards. Accordingly, we formulate the problem of selecting clients with unknown data quality as a CMAB model. Each client represents an arm in this formulation, and its data

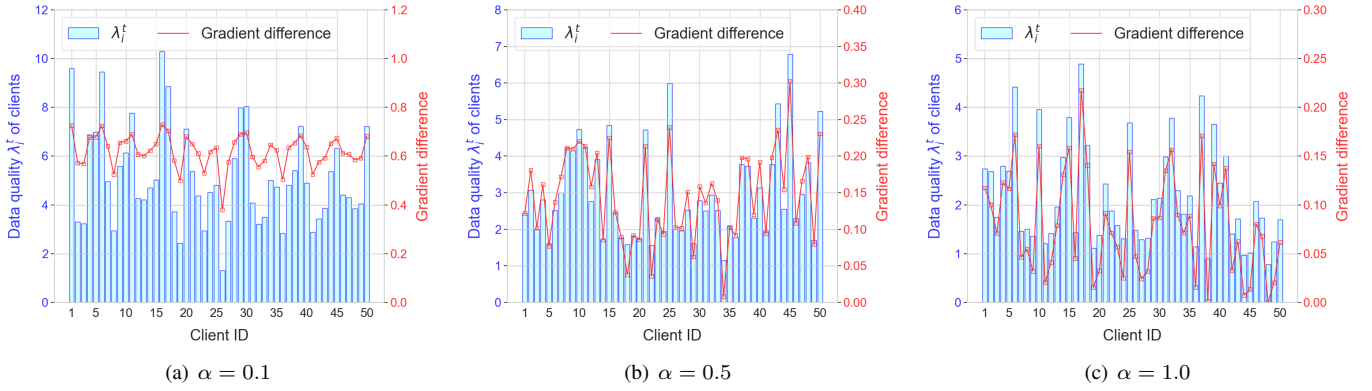


Fig. 2. Correspondence between λ_i and gradient differences of each client on the MNIST dataset with different α .

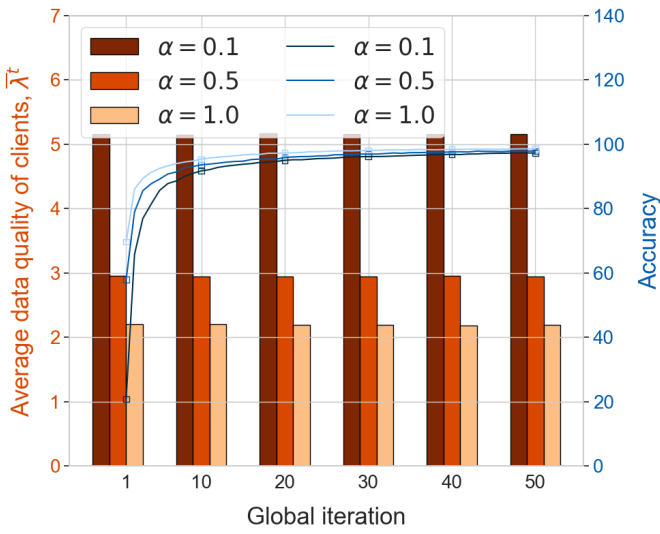


Fig. 3. Accuracy variation and average lambda on the MNIST dataset under different α .

quality corresponds to the reward it receives. The aggregator selects K clients and learns their data qualities in each global iteration.

Based on the above modeling, the client selection problem with unknown data quality can be formulated as an online learning problem to the expected cumulative gradient difference, which is shown as follows:

$$\min_{\mathcal{W}_t} \mathbb{E} \left[\sum_{t \in \mathcal{T}} \sum_{w_i \in \mathcal{W}_t} \lambda_i^t \right] \text{ s.t. } |\mathcal{W}_t| = K, \forall t \in \mathcal{T}. \quad (7)$$

C. Utility Function

During each global iteration, the aggregator disburses rewards to clients to incentivize their data contributions. These rewards are contingent upon the unit reward set by the aggregator for the clients' data. The total reward a client receives in the t -th global iteration is determined by both the amount of data they contribute and the quality of that data. On the other hand, clients decide the amount of data they contribute to the FL training to maximize their profit. This interaction between the aggregator and clients is naturally modeled as a two-stage Stackelberg game. In this framework, for the t -th global

iteration, the aggregator, acting as the leader, first sets the unit reward for clients' data, while the selected clients in $\mathcal{W}_t$, as followers, respond by determining their data contribution strategies based on the leader's reward.

The unit data reward provided by the aggregator is represented by τ^t . The final reward received by client w_i in t -th global iteration is further adjusted based on the quality of its data. Specifically, let λ represent the maximum gradient difference the aggregator tolerated. λ is a policy threshold set by the aggregator, which can be determined empirically based on specific task requirements or even adjusted dynamically. Then, the total reward for the client w_i in the t -th global iteration is as follows:

$$r_i^t = \begin{cases} \tau^t d_i^t \left(1 - \frac{\lambda_i^t}{\lambda}\right), & \text{if } \lambda_i^t < \lambda, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Our current model assumes that clients act as rational utility maximizers without engaging in complex strategic behaviors, such as collusion or misrepresentation of their capabilities.

The data-computational cost of client w_i is given by $a_i(d_i^t)^2$ [49], where $a_i > 0$ is the cost factor of client w_i . Consequently, the utility of client w_i in the t -th global iteration can be calculated as:

$$U_i^t = \begin{cases} r_i^t - a_i(d_i^t)^2, & \text{if } w_i \in \mathcal{W}_t, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

The aggregator's cost consists of the training performance (e.g., model accuracy loss) and the total reward paid to the participating clients. The cost of the aggregator in the t -th global iteration can be expressed as:

$$C_a^t = \nu_1 A_{loss}^t + \nu_2 R^t, \quad (10)$$

where ν_1 and ν_2 balance model accuracy loss A_{loss}^t and total reward R^t paid to the clients in the t -th global iteration.

To quantify the loss in model accuracy, we derive a convergence bound for the difference between the current training loss and the optimal loss. This convergence bound serves as a criterion for describing the model's accuracy loss. For the convergence analysis, we make the following assumptions regarding the loss function $F_i(\cdot)$ of client w_i [48].

Assumption 1. The global loss function $F(\cdot)$ is defined

as $F(\cdot) = \sum_{w_i \in \mathcal{W}^t} \left(\frac{d_i^t}{D^t} F_i(\cdot) \right)$. For any θ in each global iteration, the following assumptions are made for both the global and local loss functions:

- (1) $F_i(\theta)$ is convex;
- (2) $F_i(\theta)$ satisfies Polyak-Lojasiewicz condition with parameter μ , which implies that $F_i(\theta) - F_i(\theta^*) \leq \frac{1}{2\mu} \|\nabla F_i(\theta)\|^2$ where θ^* is the optimal model parameter;
- (3) $F(\theta^0) - F(\theta^*) = \zeta$;
- (4) $F_i(\theta)$ is β -Lipschitz, i.e., $\|F_i(\theta) - F_i(\theta')\| \leq \beta \|\theta - \theta'\|$ for any θ, θ' ;
- (5) $F_i(\theta)$ is ρ -Lipschitz smooth, i.e., $\|\nabla F_i(\theta) - \nabla F_i(\theta')\| \leq \rho \|\theta - \theta'\|$ for any θ, θ' , where ρ is a constant determined by the practical loss function;
- (6) $F_i(\theta)$ is an unbiased estimate of the full data gradient, i.e., $\mathbb{E}[\nabla F_i(\theta; b_i^t)] = \nabla F_i(\theta)$;
- (7) $F_i(\theta)$ is gradient variance bounded, i.e., $\|\nabla F_i(\theta; b_i^t) - \nabla F_i(\theta)\|^2 \leq \frac{\psi^2}{b_i^t}$.

It is noteworthy that these assumptions, particularly convexity and the Polyak-Lojasiewicz condition, are often violated in deep learning applications like those using ResNet-9. The loss of deep neural networks is highly non-convex, featuring numerous local minima and saddle points. In practice, violating these assumptions implies that while our algorithm cannot guarantee finding the single global optimum, it converges to a high-quality local solution. The primary value of our convexity-based analysis, therefore, is to provide a tractable framework for understanding the FL system. Specifically, our theoretical results provide crucial insights into how system parameters influence the rate and stability of convergence, a trend that persists even in non-convex settings. Furthermore, our approach is consistent with the goals of many analyses in non-convex FL [50], which aim to show convergence to a stationary point (i.e., where the expected gradient norm approaches zero) rather than a global minimum.

Theorem 1. *Based on the Assumption 1, in the t -th global iteration, by setting $\eta < 1/\rho$, the convergence upper bound is given as:*

$$F(\theta^{t+1}) - F(\theta^*) \leq \varphi^{tH} \zeta + (1 - \varphi^{tH}) \varrho \Lambda(\mathcal{W}), \quad (11)$$

where $\varphi = \rho\mu\eta^2 - 2\mu\eta + 1$, $\varrho = \frac{\beta[(\rho\eta+1)^{H-1}-1]}{\rho(1-\varphi^{tH})}$, $\Lambda(\mathcal{W}) = \sum_{w_i \in \mathcal{W}} \left[\frac{d_i^t}{\sum_{w_j \in \mathcal{W}} d_j^t} \left(-\frac{\psi}{\sqrt{b_i^t}} + \lambda_i^t \right) \right]$.

Proof. See Appendix A. $\square$

Based on Theorem 1, we use the convergence upper bound to denote the model accuracy loss:

$$A_{loss}^t = \varphi^{tH} \zeta + (1 - \varphi^{tH}) \varrho \Lambda(\mathcal{W}_t). \quad (12)$$

Thus, the total reward R^t paid by the aggregator to $\mathcal{W}_t$ in the t -th global iteration can be calculated as follows:

$$R^t = \sum_{w_i \in \mathcal{W}_t} r_i^t. \quad (13)$$

D. Problem Formulation

The interaction between the aggregator and the clients occurs in two stages. In the first stage, the aggregator provides

a uniform unit data reward. In the second stage, clients determine their data contribution strategies to maximize utility based on the provided unit data reward. Next, we define the problems for each stage in the t -th global iteration.

Problem 1. *In the first stage, the aggregator minimizes costs by adjusting the uniform unit data reward:*

$$\min_{\tau^t} C_a^t \text{ s.t. } \tau^t \geq 0. \quad (14)$$

Problem 2. *In the second stage, the client w_i in $\mathcal{W}_t$ maximizes utility by adjusting the data contribution strategy:*

$$\max_{d_i^t} U_i^t \text{ s.t. } 0 \leq d_i^t \leq D_i \quad (15)$$

In this Stackelberg game, the aggregator, as the leader, determines an optimal unit data reward $\tau^{t,*}$, ensuring that once each client has determined its optimal strategy, the aggregator cannot reduce its cost by adopting any alternative strategy τ^t . The clients, as followers, specify their optimal data contribution strategies, $d_i^{t,*}$, based on the given unit data reward, τ^t . In summary, our goal is to determine the Stackelberg Equilibrium (SE) point of the game, where neither of the players in the game wants to change their strategy unilaterally. The SE point in our model is defined as follows.

Definition 1. *Let $\mathbf{d}^{t,*} = \{d_i^{t,*} | w_i \in \mathcal{W}_t\}$ denote the set of strategies of all of the participating clients in the t -th global iteration. The point $(\tau^{t,*}, \mathbf{d}^{t,*})$ is the Stackelberg equilibrium point if the following conditions can be satisfied:*

$$C_a^t(\tau^{t,*}, \mathbf{d}^{t,*}) \leq C_a^t(\tau^t, \mathbf{d}^{t,*}), \quad (16)$$

$$U_i^t(\tau^{t,*}, \mathbf{d}^{t,*}) \geq U_i^t(\tau^{t,*}, \{d_i^t, \mathbf{d}_{-i}^{t,*}\}), \forall w_i \in \mathcal{W}_t. \quad (17)$$

where τ^t and d_i^t are arbitrary feasible strategies for the aggregator and the client $w_i \in \mathcal{W}_t$, respectively.

IV. INCENTIVE MECHANISM DESIGN

In this section, we introduce an Incentive Mechanism based on the CMAB framework and the Stackelberg game (IMCS). First, we design a client selection scheme leveraging the CUCB strategy. Then, we determine the SE point using the backward induction method based on the selected client set.

A. Client Selection with Unknown Data Quality

The aggregator must explore some suboptimal clients to identify potential optimal clients, given the uncertainty in client data quality. Simultaneously, the aggregator can select the current best clients based on learned results. The former process is referred to as exploration, while the latter is known as exploitation. To balance the trade-off between exploration and exploitation, we propose a CUCB-based data quality index as the selection criterion, which is detailed as follows.

When client w_i is selected in the t -th global iteration (i.e., $w_i \in \mathcal{W}_t$), w_i contributes data of size d_i^t . Thus, every time client w_i participates in FL, the aggregator learns its λ_i^t . Denoted by n_i^t the selected times of client w_i until the t -th global iteration, we have

$$n_i^t = n_i^{t-1} + \mathbb{I}\{w_i \in \mathcal{W}_t\}, \quad (18)$$

where $n_i^0 = 0$. Then, we introduce $\tilde{\lambda}_i^t$ to represent the learned data quality of client w_i until the t -th global iteration:

$$\tilde{\lambda}_i^t = \frac{\tilde{\lambda}_i^{t-1} n_i^{t-1} + \lambda_i^t \mathbb{I}\{w_i \in \mathcal{W}_t\}}{n_i^t}, \quad (19)$$

where $\tilde{\lambda}_i^0 = 0$.

The data quality of clients, learned during the exploration phase, is utilized for client selection in the exploitation phase. Instead of directly using the learned data quality, we define a CUCB-based data quality index, denoted as $\hat{\lambda}_i^t$, which accounts for the uncertainty in the estimation. Specifically, the learned data quality may involve some level of error. Thus, we introduce the upper confidence bound ε_i^t to guarantee that clients selected less frequently are given more opportunities for selection. ε_i^t is based on a straightforward heuristic called optimism in the face of uncertainty. The CUCB-based data quality $\hat{\lambda}_i^t$ for client w_i is defined as:

$$\hat{\lambda}_i^t = \tilde{\lambda}_i^t - \varepsilon_i^t, \quad \varepsilon_i^t = \sqrt{\frac{(K+1) \ln(\sum_{w_j \in \mathcal{W}} n_j^t)}{n_i^t}}. \quad (20)$$

In the first global iteration, all clients are selected to guarantee their initial participation in FL. After this initialization phase, during each subsequent global iteration, the top K clients with the lowest CUCB-based data quality are chosen to participate in FL.

B. Solution of the Stackelberg Game

We employ the backward induction method to analyze the Stackelberg game in the t -th global iteration. First, we derive the optimal data contribution $d_i^{t,*}$ of client $w_i \in \mathcal{W}_t$ under a given uniform unit data reward τ^t . Next, we derive the aggregator's optimal unit data reward $\tau^{t,*}$. In addition, we prove that the SE point $(\tau^{t,*}, d^{t,*})$ is unique.

First, we analyze the client w_i 's optimal strategy.

Theorem 2. *Given any uniform unit data reward τ^t , the optimal data contribution strategy $d_i^{t,\circ}$ of the client w_i can be determined by:*

$$d_i^{t,\circ} = \begin{cases} 0, & \text{if } \tau^t = 0, \\ D_i, & \text{if } \tau^t \geq \frac{2a_i D_i}{1 - \lambda_i^t / \lambda}, \\ \frac{\tau^t (1 - \lambda_i^t / \lambda)}{2a_i}, & \text{otherwise.} \end{cases} \quad (21)$$

Proof. We calculate the first-order and second-order derivatives of U_i^t with respect to d_i^t as follows:

$$\frac{\partial U_i^t}{\partial d_i^t} = \tau^t \left(1 - \frac{\lambda_i^t}{\lambda}\right) - 2a_i d_i^t. \quad (22)$$

$$\frac{\partial^2 U_i^t}{\partial (d_i^t)^2} = -2a_i < 0. \quad (23)$$

Therefore, the utility function of client w_i is strictly concave, and the problem for each client w_i is a concave optimization problem. Accordingly, we employ the Karush-Kuhn-Tucker (KKT) conditions to solve this problem.

Let γ_1, γ_2 be the Lagrange multipliers associated with the aggregator's cost function (10). Then, the Lagrangian function

is defined as follows:

$$L_i^t = U_i^t + \gamma_1 (D_i - d_i^t) + \gamma_2 d_i^t. \quad (24)$$

The KKT conditions for the follower (client w_i) problem 2 are listed as follows.

1) Stationarity conditions:

$$\frac{\partial L_i^t}{\partial d_i^t} = \tau^t \left(1 - \frac{\lambda_i^t}{\lambda}\right) - 2a_i d_i^t - \gamma_1 + \gamma_2 = 0. \quad (25)$$

2) Primal feasibility condition:

$$D_i - d_i^t \geq 0, d_i^t \geq 0. \quad (26)$$

3) Dual feasibility conditions:

$$\gamma_1, \gamma_2 \geq 0. \quad (27)$$

4) Complementary slackness conditions:

$$\gamma_1 (D_i - d_i^t) = 0, \gamma_2 d_i^t = 0. \quad (28)$$

The optimal solution for the problem is taken in one of the following three cases.

1) Case 1: $d_i^t = 0$. In this case, $\gamma_1 = 0$ according to KKT condition (28). Substitute it into KKT condition (25), we have $\gamma_2 = -\tau^t (1 - \lambda_i^t / \lambda)$. Due to KKT condition (27), we have the following conclusion:

$$d_i^{t,\circ} = 0, \text{ if } \tau^t = 0. \quad (29)$$

2) Case 2: $d_i^t = D_i$. In this case, $\gamma_2 = 0$ according to KKT condition (28). Substitute it into KKT condition (25), we have $\gamma_1 = \tau^t (1 - \lambda_i^t / \lambda) - 2a_i D_i$. Due to KKT condition (27), we have the following conclusion:

$$d_i^{t,\circ} = D_i, \text{ if } \tau^t \geq \frac{2a_i D_i}{1 - \lambda_i^t / \lambda}. \quad (30)$$

3) Case 3: $0 < d_i^t < D_i$. In this case, $\gamma_1 = 0$ and $\gamma_2 = 0$ according to KKT conditions (28) and (28). Substitute these into KKT condition (25), we have $\tau^t (1 - \lambda_i^t / \lambda) - 2a_i d_i^t = 0$. Then, we have the following conclusion:

$$d_i^{t,\circ} = \frac{\tau^t (1 - \lambda_i^t / \lambda)}{2a_i}. \quad (31)$$

The optimal data contribution $d_i^{t,\circ}$ of client w_i can be shown as Eqn. (21), and then this theorem is proven. $\square$

Before analyzing the aggregator's optimal strategy, we set $\xi_i^t = \frac{1 - \lambda_i^t / \lambda}{2a_i}$ for client w_i . Then, we list all the clients in $\mathcal{W}_t$ in ascending order of D_i / ξ_i^t for each client $w_i \in \mathcal{W}_t$. Let the sorted list of the selected clients $\mathcal{W}_t$ be denoted as $\mathcal{W}_t^s$. Thus, we have $\mathcal{W}_t^s = (w_{s_1}, w_{s_2}, \dots, w_{s_K})$ with $D_{s_1} / \xi_{s_1}^t \leq D_{s_2} / \xi_{s_2}^t \leq \dots \leq D_{s_K} / \xi_{s_K}^t$.

Lemma 1. *When $\tau^t \in (0, D_{s_1} / \xi_{s_1}^t)$, the unique optimal unit data reward of the aggregator can be expressed as follows:*

$$\tau_0^{t,\circ} = \left(\frac{\nu_1 (1 - \varphi^{tH}) \varrho \sum_{w_i \in \mathcal{W}_t} \frac{\psi \sqrt{\xi_i^t H}}{\sum_{w_j \in \mathcal{W}_t} \xi_j^t}}{4\nu_2 \sum_{w_j \in \mathcal{W}_t} \xi_j^t \left(1 - \frac{\lambda_j^t}{\lambda}\right)} \right)^{\frac{2}{5}}. \quad (32)$$

Proof. The data contribution strategy of each client w_i is $\frac{\tau^t(1-\lambda_i^t/\lambda)}{2a_i}$ when $\tau^t \in (0, D_{s_1}/\xi_{s_1}^t)$. Substituting then into the cost function (10) of the aggregator, we have:

$$C_a^t = \nu_1 \left[\varphi^{tH} \zeta + (1 - \varphi^{tH}) \varrho \sum_{w_i \in \mathcal{W}_t} \frac{\psi \sqrt{\xi_i^t H / \tau^t} + \xi_i^t \lambda_i^t}{\sum_{w_j \in \mathcal{W}^t} \xi_j^t} \right] + \nu_2 (\tau^t)^2 \sum_{w_i \in \mathcal{W}_t} \xi_i^t \left(1 - \frac{\lambda_i^t}{\lambda} \right). \quad (33)$$

Then, we calculate the first-order and second-order derivatives of Eqn. (33) with respect to τ^t as follows:

$$\frac{\partial C_a^t}{\partial \tau^t} = \nu_1 (1 - \varphi^{tH}) \varrho \sum_{w_i \in \mathcal{W}_t} \frac{-\psi (\tau^t)^{\frac{3}{2}} \sqrt{\xi_i^t H}}{2 \sum_{w_j \in \mathcal{W}_t} \xi_j^t} + 2\nu_2 \tau^t \sum_{w_i \in \mathcal{W}^t} \xi_i^t \left(1 - \frac{\lambda_i^t}{\lambda} \right), \quad (34)$$

$$\frac{\partial^2 C_a^t}{\partial (\tau^t)^2} = \nu_1 (1 - \varphi^{tH}) \varrho \sum_{w_i \in \mathcal{W}_t} \frac{3\psi (\tau^t)^{\frac{5}{2}} \sqrt{\xi_i^t H}}{4 \sum_{w_j \in \mathcal{W}_t} \xi_j^t} + 2\nu_2 \sum_{w_i \in \mathcal{W}^t} \xi_i^t \left(1 - \frac{\lambda_i^t}{\lambda} \right). \quad (35)$$

In practical applications, we can ensure $1 - \varphi^{tH} > 0$ by appropriately choosing the values of ρ , η , and μ . As a result, the second derivative of C_a^t is strictly positive, indicating that the cost function C_a^t of the aggregator is strictly convex when $\tau^t \in (0, D_{s_1}/\xi_{s_1}^t)$. Thus, the unique optimal strategy can be obtained by solving $\frac{\partial C_a^t}{\partial \tau^t} = 0$, and the optimal unit data reward $\tau_0^{t,\circ}$, for $\tau^t \in (0, D_{s_1}/\xi_{s_1}^t)$, is derived as Eqn. (32). $\square$

Next, we define $\sigma_D^{k-} = \sum_{i=1}^k D_{s_i}$ as the sum of D_{s_i} of the first k clients in $\mathcal{W}_t^s$ and $\sigma_\xi^{(k+1)+} = \sum_{i=k+1}^K \xi_{s_i}^t$ as the sum of $\xi_{s_i}^t$ of the last $K-k$ clients in $\mathcal{W}_t^s$. To ensure that the optimal data contribution of each client in $\mathcal{W}_t^s$ is bounded within the Stackelberg game, we make the following assumption:

Assumption 2. For any client $w_{s_k} \in \mathcal{W}_t^s$, there is

$$D_{s_k} / \xi_{s_k}^t \leq (1 + \sqrt{2}/3) \sigma_D^{k-} / \sigma_\xi^{(k+1)+}. \quad (36)$$

Building on the above sufficient assumption, we can guarantee the existence and uniqueness of an SE for the aggregator in the game as follows:

Lemma 2. Under Assumption 2, the optimal unit data reward $\tau_k^{t,\circ}$ of the aggregator exists and is unique when $\tau^t \in [D_{s_k} / \xi_{s_k}^t, D_{s_{k+1}} / \xi_{s_{k+1}}^t]$ for $\forall w_{s_k} \in \mathcal{W}_t^s$.

Proof. According to Eqn. (21), the data contribution strategy for the first k clients in $\mathcal{W}_t^s$ contributes their total data (i.e., D_i), and the remaining $K-k$ clients in $\mathcal{W}_t^s$ contributes $\frac{\tau^t(1-\lambda_i^t/\lambda)}{2a_i}$ when $\tau^t \in [D_{s_k} / \xi_{s_k}^t, D_{s_{k+1}} / \xi_{s_{k+1}}^t]$. Substituting then into the cost function (10) of the aggregator, we have:

$$C_a^t = \nu_1 \varphi^{tH} \zeta + \nu_1 (1 - \varphi^{tH}) \varrho \left[\sum_{i=1}^k \frac{\psi \sqrt{D_{s_i} H} + D_{s_i} \lambda_{s_i}^t}{\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+}} + \sum_{i=k+1}^K \frac{\psi \sqrt{\tau^t \xi_{s_i}^t H} + \tau^t \xi_{s_i}^t \lambda_{s_i}^t}{\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+}} \right]$$

$$+ \nu_2 \left(\tau^t \sum_{i=1}^k D_{s_i} \left(1 - \frac{\lambda_{s_i}^t}{\lambda} \right) + (\tau^t)^2 \sum_{i=k+1}^K \xi_{s_i}^t \left(1 - \frac{\lambda_{s_i}^t}{\lambda} \right) \right). \quad (37)$$

Then, we calculate the first-order and second-order derivatives of Eqn. (37) with respect to τ^t as follows:

$$\frac{\partial C_a^t}{\partial \tau^t} = \nu_1 (1 - \varphi^{tH}) \varrho \left[\sum_{i=1}^k \frac{-\xi_{s_i}^t (\psi \sqrt{D_{s_i} H} + D_{s_i} \lambda_{s_i}^t)}{(\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+})^2} + \sum_{i=k+1}^K \frac{\psi \sqrt{\xi_{s_i}^t H} (\frac{\sigma_D^{k-}}{\sqrt{\tau^t}} + \sigma_\xi^{(k+1)+} \sqrt{\tau^t}) + 2\xi_{s_i}^t \lambda_{s_i}^t \sigma_D^{k-}}{2(\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+})^2} \right] + \nu_2 \left(\sum_{i=1}^k D_{s_i} \left(1 - \frac{\lambda_{s_i}^t}{\lambda} \right) + 2\tau^t \sum_{i=k+1}^K \xi_{s_i}^t \left(1 - \frac{\lambda_{s_i}^t}{\lambda} \right) \right), \quad (38)$$

$$\frac{\partial^2 C_a^t}{\partial (\tau^t)^2} = \nu_1 (1 - \varphi^{tH}) \varrho \left[\sum_{i=1}^k \frac{2(\xi_{s_i}^t)^2 (\psi \sqrt{D_{s_i} H} + D_{s_i} \lambda_{s_i}^t)}{(\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+})^3} + \sum_{i=k+1}^K \frac{\psi \sqrt{\xi_{s_i}^t H} G(\tau^t)}{4(\sigma_D^{k-} + \tau^t \sigma_\xi^{(k+1)+})^3} \right] + 2\nu_2 \sum_{i=k+1}^K \xi_{s_i}^t \left(1 - \frac{\lambda_{s_i}^t}{\lambda} \right). \quad (39)$$

where $G(\tau^t) = 3(\sigma_\xi^{(k+1)+})^2 (\tau^t)^{\frac{1}{2}} - 6\sigma_D^{k-} \sigma_\xi^{(k+1)+} (\tau^t)^{-\frac{1}{2}} - (\sigma_D^{k-})^2 (\tau^t)^{-\frac{3}{2}}$.

In the second-order (39) of C_a^t , the first and third items are positive. However, the positivity of the second term requires further discussion: All components are positive except for $G(\tau^t)$ in the second item. The second derivative of $G(\tau^t)$ with respect to τ^t is always negative, and the first derivative is always positive, so $G(\tau^t)$ increases monotonically when $\tau^t > 0$. Moreover, since $\lim_{\tau^t \rightarrow 0^+} G(\tau^t) = -\infty$ and $\lim_{\tau^t \rightarrow +\infty} G(\tau^t) = +\infty$, along with the application of the Zero-point theorem, $G(\tau^t)$ has a unique zero point, which is given by $\tau^t = (1 + 2/\sqrt{3}) \sigma_D^{k+} / \sigma_\xi^{(k+1)+}$. Thus, $G(\tau^t)$ is positive when $\tau^t > (1 + 2/\sqrt{3}) \sigma_D^{k+} / \sigma_\xi^{(k+1)+}$. Under Assumption 2, $G(\tau^t)$ is always positive when $\tau^t \in [D_{s_k} / \xi_{s_k}^t, D_{s_{k+1}} / \xi_{s_{k+1}}^t]$.

Based on the above analysis, the cost function C_a^t of the aggregator is strictly convex when $\tau^t \in [D_{s_k} / \xi_{s_k}^t, D_{s_{k+1}} / \xi_{s_{k+1}}^t]$ under Assumption 2. The optimal unit data reward can be obtained by solving $\frac{\partial C_a^t}{\partial \tau^t} = 0$. However, due to the extremely complex expression of $\frac{\partial C_a^t}{\partial \tau^t}$, it is difficult to derive an analytical solution for the optimal strategy. Therefore, we will use Newton's Method [51] to compute its numerical solution and denote the optimal strategy as $\tau_k^{t,\circ}$. Thus, the proof is complete. $\square$

Notably, the optimal unit data reward $\tau_K^{t,\circ}$ is $D_{s_K} / \xi_{s_K}^t$ when $\tau^t \geq D_{s_K} / \xi_{s_K}^t$. This is because when $\tau^t \geq D_{s_K} / \xi_{s_K}^t$, increasing τ^t does not increase the data contribution of each client, thus no longer reducing C_a^t . Combining Lemma 1 and

Algorithm 1: Client Selection

```

input :  $t, \mathcal{W}, K$ .
output: Selected client set  $\mathcal{W}_t$ .
1 if  $t == 1$  then
2   return  $\mathcal{W}$ ; // Explore all clients
   initially
3 else
4    $\mathcal{W}_t \leftarrow$  top- $K$  clients with the smallest  $\hat{\lambda}_i^{t-1}$ ;
5   for  $w_i \in \mathcal{W}$  do
6     Calculate CUCB index  $\hat{\lambda}_i^t$  by Eqn. (18)-(20);
7   return  $\mathcal{W}_t$ ;

```

Lemma 2, we have the following theorem about the optimal unit data reward of the aggregator.

Theorem 3. Let $\tau^{t,\circ} = \{\tau_0^{t,\circ}, \tau_1^{t,\circ}, \dots, \tau_K^{t,\circ}\}$, the optimal unit data reward of the aggregator can be expressed as follows:

$$\tau^{t,*} = \arg \min_{\tau^{t,\circ}} C_a^t. \quad (40)$$

Proof. Lemma 1 and Lemma 2 derive the optimal unit data reward when $\tau^t \in (0, D_{s_1}/\xi_{s_1}^t)$ and $\tau^t \in [D_{s_k}/\xi_{s_k}^t, D_{s_{k+1}}/\xi_{s_{k+1}}^t)$, for any $k = 1, \dots, K-1$. Combining with $\tau_K^{t,\circ}$, the optimal strategy over the entire range of τ^t can be obtained by Eqn. (40). Thus, the proof is complete. $\square$

Next, we will prove that the Stackelberg equilibrium of the game between the aggregator and clients exists and is unique.

Theorem 4. The SE point of the game between the aggregator and clients exists and is unique.

Proof. As analyzed in the proof of Theorem 2, each client will find its optimal strategy in one of the three cases. This implies that, after the aggregator provides the unit data reward strategy, the strategy for each rational client is unique. Based on Lemma 1, Lemma 2, and Theorem 3, the aggregator's optimal strategy is unique. Therefore, we can conclude that the SE point between the aggregator and the clients exists and is unique. $\square$

C. Algorithm Design

The incentive mechanism we proposed consists of two components: selecting clients with unknown data quality and solving the SE point. Next, we first introduce the client selection algorithm for unknown data quality, and then, based on this selection algorithm, we present the IMCS algorithm.

For Algorithm 1, we initialize the selected client set as empty in line 1. If it is the first global iteration, to explore all clients' data quality, we select all clients to participate in FL (Line 2), i.e., $\mathcal{W}_t = \mathcal{W}$. For subsequent iterations, we select the smallest K clients of CUCB-based data quality into $\mathcal{W}_t$ in line 4. In lines 5-6, we calculate the parameters for updating CUCB-based data quality for each client.

Algorithm 2 calculates the optimal strategy for the aggregator and the selected clients $\mathcal{W}_t$ by solving the SE point within each global iteration (Lines 1-9). Specifically, we first

Algorithm 2: Incentive Mechanism (IMCS)

```

input :  $\mathcal{T}, \mathcal{W}, K$ .
output: Optimal strategy  $d^{t,*}$  and  $\tau^{t,*}$  for  $t = 1 \dots T$ .
1 for  $t \in \mathcal{T}$  do
2   Get  $\mathcal{W}_t$  by Algorithm 1 ( $t, \mathcal{W}, K$ );
3   Initialize  $\tau_{now}^t = 0, d_{now}^t$ ;
4   repeat
5      $\tau_{last}^t \leftarrow \tau_{now}^t, d_{last}^t \leftarrow d_{now}^t$ ;
6     Calculate  $d_{now}^t$  based on  $\tau_{last}^t$  by Eqn. (21);
7     Calculate  $\tau_{now}^t$  based on  $d_{now}^t$  by Eqn. (40);
8   until Convergence or  $iter > iter_{max}$ ;
9    $d^{t,*} = d_{now}^t, \tau^{t,*} = \tau_{now}^t$ ;
10  Output:  $d^{t,*}, \tau^{t,*}$ ;

```

obtain the selected clients set $\mathcal{W}_t$ using Algorithm 1 in the t -th global iteration (Line 2). After initializing the variables for the Stackelberg game iteration in line 3, we iteratively solve for the optimal strategies $d^{t,*}$ and $\tau^{t,*}$ during the t -th global iteration (Lines 4-9). The loop condition is that the current optimal strategy differs from the optimal strategy of the previous Stackelberg game iteration, and the number of iterations $iter$ has not reached the maximum limit $iter_{max}$.

For Algorithm 1, since parameters for all clients need to be updated in lines 5-6, the computational complexity is $\mathcal{O}(N)$, regardless of whether it is the first global iteration. For Algorithm 2, the incentive mechanism returns the optimal strategies: $\{d^t, \tau^t | t \in \mathcal{T}\}$ after T global iterations. The computational complexity of solving the follower's optimal strategy is $\mathcal{O}(K)$, and that of solving the leader's optimal strategy is $\mathcal{O}(K \log(\log(1/\epsilon)))$ [51], where ϵ is the target accuracy of Newton's Method. Therefore, the theoretical computational complexity of Algorithm 2 is $\mathcal{O}(T(N + SK \log(\log(1/\epsilon))))$. It is worth mentioning that, in practical performance evaluations, the number of Stackelberg game iterations typically does not exceed 5. Thus, the practical computational complexity of Algorithm 2 is approximately $\mathcal{O}(T(N + K \log(\log(1/\epsilon))))$. Furthermore, while the theoretical complexity depends on the total number of clients N , it is crucial to note that in most practical FL deployments, the communication overhead for transmitting models far exceeds the server-side computation time. Therefore, the computational cost of our mechanism is acceptable.

V. REGRET ANALYSIS

The primary difficulty in applying CUCB to address the CMAB problem lies in maintaining a balance between exploration and exploitation. Specifically, this involves striking a balance between seeking out potential clients with high data quality and prioritizing the selection of those clients with high data quality who have already been thoroughly explored. A common way to evaluate this balance is through the concept of regret, which represents the loss in data quality caused by not consistently selecting the clients with the highest global data quality throughout all global iterations. Before defining the regret, we first introduce the following related variables. In each global iteration, there are $C_N^K = \frac{N!}{K!(N-K)!}$ possible

combinations of $\mathcal{W}_t$ that can be selected. Let Ω denote the set of all possible $\mathcal{W}_t$. For each $\mathcal{W}_j$ in Ω , its expected total data quality is $\lambda_{\mathcal{W}_j} = \sum_{w_i \in \mathcal{W}_j} \lambda_i$. Additionally, we denote $\mathcal{W}_*$ as the client set with the lowest total data quality in Ω . The difference in the expected total data quality achieved by selecting $\mathcal{W}_*$ and $\mathcal{W}_j$ is expressed as follows:

$$\Delta_j = \sum_{w_i \in \mathcal{W}_j} \lambda_i - \sum_{w_i \in \mathcal{W}_*} \lambda_i. \quad (41)$$

For any $\mathcal{W}_j \in \Omega$, we represent the number of selection of $\mathcal{W}_j$ until t -th global iteration as ζ_j^t . Then, we have the following definition of regret:

Definition 2. *The expected data quality regret of client selection by using CUCB can be calculated as follows:*

$$\mathcal{R}(T) = \sum_{\mathcal{W}_j \in \Omega} \Delta_j \mathbb{E}[\zeta_j^T], \quad (42)$$

To ensure that the expected data quality regret $\mathcal{R}$ is bounded, we make the following assumptions:

Assumption 3. *For the number of clients selected for each global iteration K , the upper bound λ_{max} and lower bound λ_{min} of data quality for each client, there is*

$$K \geq 2(\lambda_{max} - \lambda_{min})^2 - 1. \quad (43)$$

Building on the above sufficient assumption, we have the following theorem about the upper bound of the regret.

Theorem 5. *The upper bound of the expected data quality regret after T global iterations is as follows:*

$$\mathcal{R}(T) \leq |\Omega|K(\lambda_{max} - \lambda_{min}) \left(\frac{4K^2(K+1)\ln(NT)}{(\Delta_{min})^2} + \frac{K\pi^2(N)^{-\frac{2(K+1)}{(\lambda_{max}-\lambda_{min})^2}} + 3}{3} \right) \quad (44)$$

Proof. See Appendix B. $\square$

VI. PERFORMANCE EVALUATION

In this section, we first describe the datasets and models used in the experiments presented in Section VI-A. Next, we introduce the baselines for performance comparison in Section VI-B. Finally, we give the simulation settings and results in Section VI-C and VI-D.

A. Datasets and Models

Dataset. We conduct extensive experiments on two real-world datasets: MNIST (<http://yann.lecun.com/exdb/mnist/>) and CIFAR-10 (<https://www.cs.toronto.edu/~kriz/cifar.html>). Specifically, the MNIST dataset [52] contains a training set of 60,000 samples and a test set of 10,000 samples. Each sample in MNIST is a 28×28 grayscale image of a hand-written digit from the numbers 0 to 9. CIFAR-10 contains 60,000 32×32 color images labeled in 10 classes with 50,000 samples for training and 10,000 samples for testing. To verify the effectiveness of IMCS in highly non-IID data scenarios, we adopt the Dirichlet distribution to divide the dataset and simulate non-IID data distributions of different

degrees [53]. Specifically, we set the parameters α of the Dirichlet distribution to 0.1, 0.5, and 1.0, respectively. The smaller the value of α , the higher the non-IID degree of the data distribution. When $\alpha = 0.1$, the client's data distribution is highly heterogeneous. Each client typically has samples from a few categories to simulate extreme non-IID scenarios. When $\alpha = 0.5$, the heterogeneity of the data distribution is moderate. There is still a certain deviation in the category distribution of the client, but it is more balanced compared with $\alpha = 0.1$. When $\alpha = 1.0$, the data distribution is close to a uniform distribution, and the sample category distribution of the client is relatively balanced, representing a lower degree of non-IID. Through this division method, we can systematically evaluate the performance of the IMCS under varying degrees of non-IID.

Models. We employ two distinct network models with varying architectures to evaluate the performance of our incentive mechanism on the aforementioned real-world datasets. For the MNIST dataset, we utilize a standard CNN model consisting of two 5×5 convolutional layers, a fully connected layer with 512 units, and a softmax output layer with 10 units [45]. For the CIFAR-10 dataset, we adopt the widely recognized ResNet-9 model [54].

B. Baselines

Since existing methods address divergent problems, they are not directly applicable as baselines for fair comparison. Accordingly, we design the following custom baseline algorithms tailored to our problem.

- **Thompson:** It modifies IMCS by replacing its client selection method. This variant employs Thompson Sampling, another well-established CMAB solution strategy [55]. All other components of IMCS remain unchanged. It allows us to benchmark the effectiveness of our client selection approach.
- **Importance:** It further explores the impact of client selection within our IMCS by replacing the CMAB-based selection strategy with Importance Sampling, which selects clients based on the probability determined by the DELTAPrac sampling method for their participation in FL [56]. All other components of IMCS remain unchanged. It allows us to evaluate the performance of our CMAB-based selection strategy.
- **Heuristic:** It replaces the client selection method in IMCS with a genetic algorithm (GA)-based heuristic. In each global iteration, the GA executes 1000 evolutionary cycles to optimize client selection, without leveraging any historical performance data. It evaluates the effectiveness of incremental learning strategies for quality-aware client selection.
- **Gradient:** It retains the client selection method in IMCS but modifies the solution method for the optimal strategy of the incentive mechanism. This variant employs iterative gradient-based optimization to numerically approximate the SE point in each global iteration. All other components of IMCS remain unchanged. It enables the evaluation of solution accuracy between theoretical

derivation and numerical approximation for the incentive mechanism.

- **OPT:** It executes our incentive mechanism when the aggregator knows the data quality of all clients. All other components of IMCS remain unchanged.

TABLE III
SIMULATION PARAMETER SETTINGS

Dataset	ρ	β	μ	ψ	λ
MNIST	30.18	0.77	7.52	20	8
CIFAR-10	18.69	0.41	3.08	15	8

C. Simulation Setup

We simulate an FL system consisting of 50 clients and an aggregator. For the MNIST dataset, the data sizes across these 50 clients follow a uniform distribution with a mean of 5960.75 samples, ranging from a minimum of 5477 to a maximum of 6703. For CIFAR-10, the client data sizes similarly follow a uniform distribution with a mean of 10713.87 samples, ranging from 8112 to 12916. Through fine-tuning the datasets of each client, client data quality λ_i^t (under the same α and K) in the MNIST dataset follows a normal distribution, while in the CIFAR-10 dataset, it follows a uniform distribution. The implementation of model training for each client is based on the PyTorch framework [57].

Different data sets yield varying parameter values, such as ρ , μ , etc. Using a learning rate of $\eta = 0.01$ and a global iteration number of $T = 100$, we empirically estimate these parameters via a concise FL training process [58], as shown in Table III. During the simulation experiments, the number of global iterations is set to 200 for MNIST and CIFAR-10. The number K of selected clients in each round of global iteration has three values: 10, 20, and 30. The learning rate is initialized as 0.01, 0.02 for CNN on MNIST and ResNet-9 on CIFAR-10, respectively. The batch size is set as 64 for CNN and 128 for ResNet-9. The balance parameters ν_1 and ν_2 are set to 100 and 10^{-7} , respectively. The tolerance parameter ϵ_N of Newton's method used to solve the optimal strategy of the aggregator is 10^{-4} .

D. Simulation Results

1) *Accuracy Analysis:* We first compare the global model accuracy of our proposed IMCS under different α and K . The training results on MNIST and CIFAR-10 are presented in Fig. 4. Overall, a higher α leads to faster convergence of accuracy, which aligns with our preliminary experiments.

As illustrated in Fig. 4, we observe that under a high non-IID degree ($\alpha = 0.1$), increasing the number of participating FL clients generally improves global model accuracy, though with diminishing marginal returns. Specifically, as shown in Figs. 4(a) and 4(b), the accuracy gain when K increases from 20 to 30 is significantly smaller than when K increases from 10 to 20. At medium non-IID degree ($\alpha = 0.5$), the MNIST results clearly demonstrate that increasing K negatively impacts

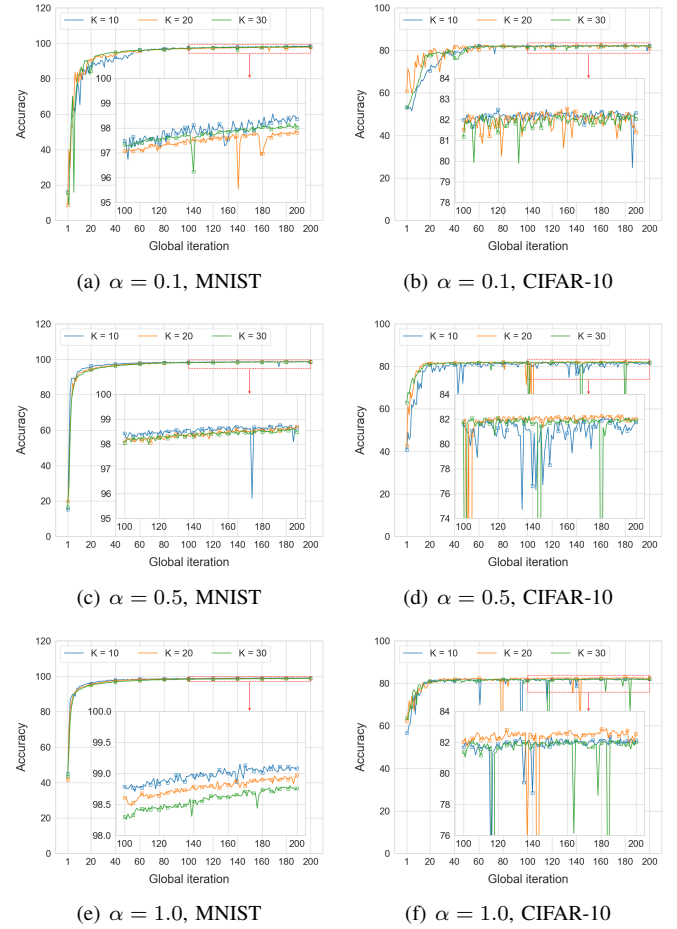


Fig. 4. Accuracy variation of IMCS for training under different α .

global model accuracy (Fig. 4(c)). Conversely, on CIFAR-10, increasing K still yields positive accuracy improvements, albeit with diminishing returns (Fig. 4(d)). Under low non-IID degree ($\alpha = 1.0$), the MNIST results resemble the medium non-IID case, showing negative impacts from increasing K (Fig. 4(e)). For CIFAR-10, however, changes in K show no discernible effect on global accuracy (Fig. 4(f)).

Next, with α fixed at 0.1, we compare our algorithm against various baselines across different K . Fig. 5 presents the global model accuracy comparisons on the MNIST and CIFAR-10 datasets, respectively. On the MNIST dataset, IMCS achieves maximum accuracy improvements of 2.59% ($K = 10$), 2.55% ($K = 20$), and 0.90% ($K = 30$) over the baselines. For CIFAR-10, the maximum improvements reach 4.33% ($K = 10$), 1.73% ($K = 20$), and 2.08% ($K = 30$). In terms of convergence rate, IMCS significantly reduces the required global iterations compared to baselines for reaching target accuracies. On MNIST to achieve 90% accuracy, it reduces global iterations by up to 15 ($K = 10$), 7 ($K = 20$), and 20 ($K = 30$); while on CIFAR-10 to reach 75% accuracy, it achieves reductions of up to 11 ($K = 10$), 12 ($K = 20$), and 24 ($K = 30$) global iterations, respectively. Overall, IMCS demonstrates superior performance compared to all baselines except OPT, especially in scenarios with a high non-ID degree.

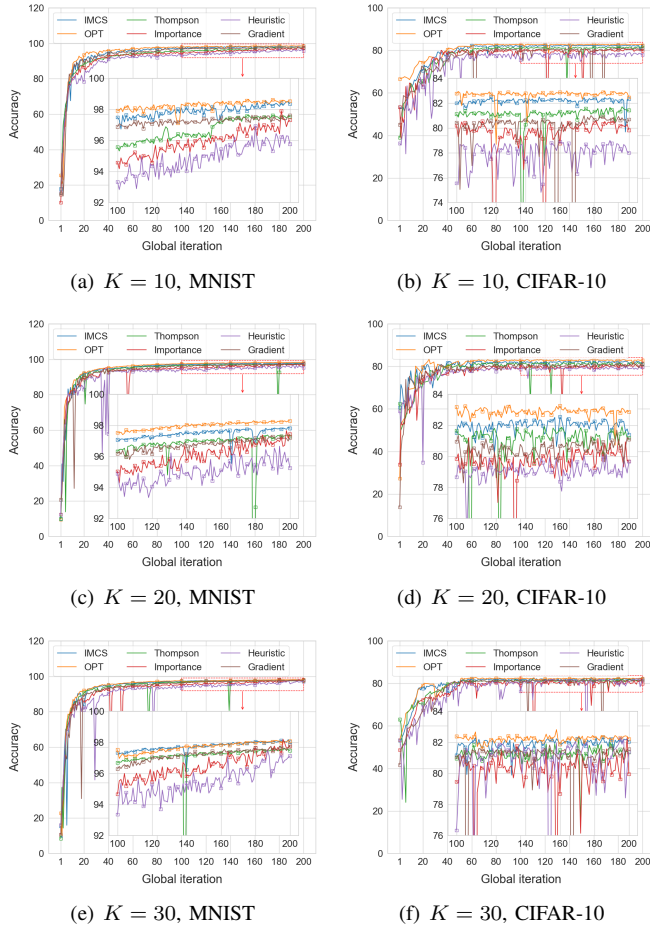


Fig. 5. Accuracy variation of each algorithm for training under different K .

TABLE IV
AVERAGE RUNNING TIME OF GLOBAL ITERATIONS (s)

α	$K = 10$		$K = 20$		$K = 30$	
	M	C	M	C	M	C
0.1	17.72	20.99	22.96	37.13	29.43	76.66
0.5	17.60	20.79	21.91	36.43	28.86	77.89
1	17.04	20.78	21.04	38.95	29.66	75.74

TABLE V
AVERAGE PEAK MEMORY OF GLOBAL ITERATIONS (MB)

α	$K = 10$		$K = 20$		$K = 30$	
	M	C	M	C	M	C
0.1	67.90	108.57	129.20	198.68	190.53	288.79
0.5	67.85	108.43	129.18	198.63	190.55	288.86
1	67.83	108.46	129.19	198.67	190.61	288.78

2) *Runtime and Memory Analysis*: To evaluate the practical overhead of the proposed IMCS mechanism, we measured the average running time and average peak memory usage for a single global iteration. The results across different datasets, non-IID degrees (α), and numbers of selected clients (K) are

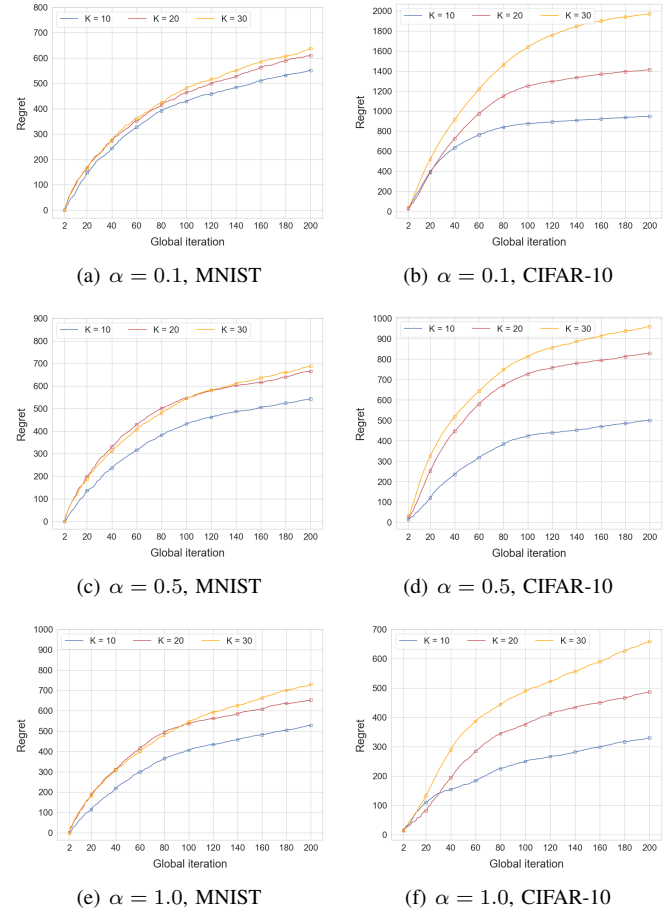


Fig. 6. Regret variation of IMCS for training under different α .

presented in Table IV and V, where M and C denote the MNIST and CIFAR-10 datasets, respectively.

The average running time is practical. On the MNIST dataset, it ranges from approximately 17.04s to 29.7s, and on the CIFAR-10 dataset, it ranges from 20.8s to 77.9s. The primary drivers of this time cost are the model complexity and the number of selected clients, which dictate the overhead of model aggregation. Notably, the runtime is not significantly affected by the degree of non-IID data distribution, demonstrating the consistent performance of our mechanism. The average peak memory usage is modest, peaking at 288.8 MB for the CIFAR-10 dataset, when $K = 30$. This low memory footprint confirms that the aggregator-side computations for solving the Stackelberg game are efficient.

Furthermore, the additional communication overhead introduced by IMCS is slight. Standard FL is bottlenecked by the exchange of large model parameters. Our mechanism only adds one minimal step: the aggregator broadcasts a single scalar value, the unit reward τ^t , to the K selected clients. This signal is trivial in size compared to the model weights, ensuring IMCS remains practical and suitable for mobile scenarios.

3) *Regret Analysis*: For Regret, we also demonstrate the performance of IMCS at different α and K , as shown in Fig. 6. The final regret ranges from 500 to 800 on the MNIST

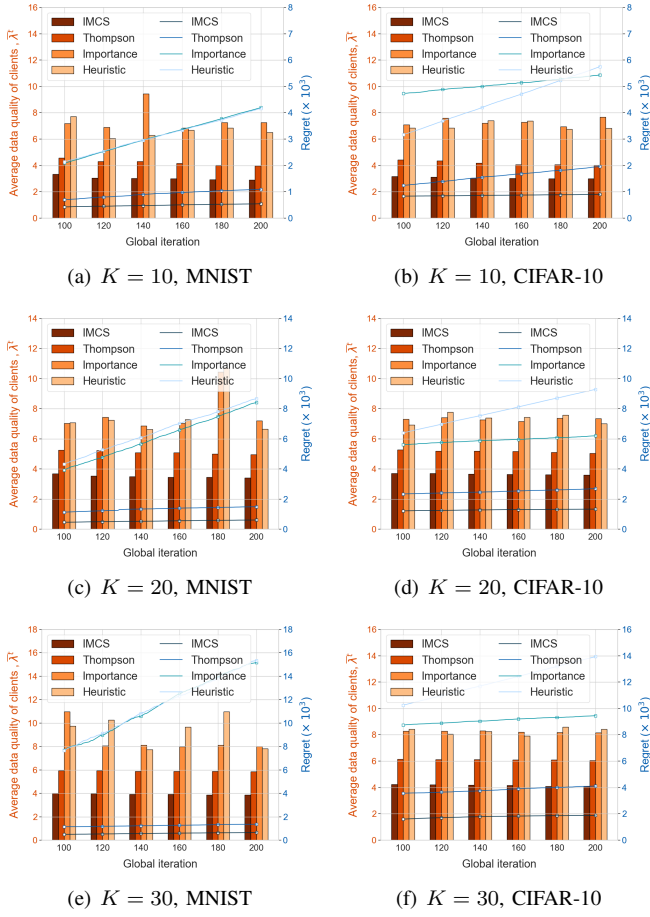


Fig. 7. Regret variation of each algorithm for training under different K .

dataset. When selecting $K = 10$ clients, the regret per epoch is the lowest. For $K = 20$ and $K = 30$, the regret for $K = 20$ is higher than for $K = 30$ in approximately the first 100 epochs under $\alpha = 0.5$ and 1.0 in Figs. 6(c) and 6(e); however, beyond this point, the regret for $K = 20$ becomes lower than that for $K = 30$. We attribute this phenomenon to the normal distribution of client data quality. Selecting $K = 10$ primarily targets clients at the distribution tails (higher quality), where quality differences are more pronounced, allowing the algorithm to identify the optimal clients faster. In contrast, selecting $K = 20$ or $K = 30$ necessitates including clients from the central region (medium quality). These clients closely cluster in terms of quality values, making them harder to distinguish. Consequently, IMCS requires longer exploration time to effectively discern their relative quality, leading to slower regret reduction and the observed crossover behavior. In the CIFAR-10 dataset, regret is lowest for $K = 10$, intermediate for $K = 20$, and highest for $K = 30$ as shown in Figs. 6(b), 6(d), and 6(f). This phenomenon can be explained by the fact that clients' data quality values are evenly spread across the entire range. Selecting a smaller K (e.g., $K = 10$) allows IMCS to concentrate precisely on the subset of clients with the highest quality. As K increases ($K = 20$, $K = 30$), the algorithm is forced to include clients with progressively lower relative quality. The inclusion of these lower-quality

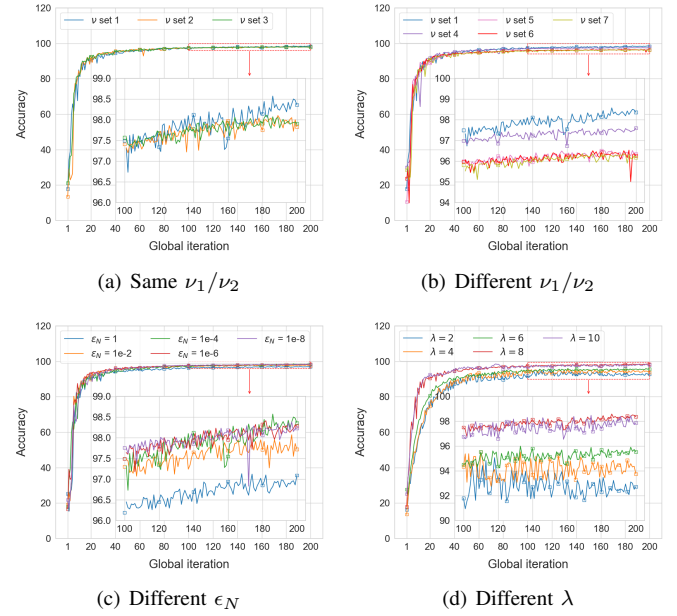


Fig. 8. Accuracy variation of IMCS for training on the MNIST dataset under different ν_1 , ν_2 , ϵ_N , and λ .

clients directly contributes to a degradation in the overall learning performance, manifesting as an increase in regret ($K = 10 < K = 20 < K = 30$).

Next, we compare the client selection effectiveness of IMCS with other baselines, as shown in Figs. 7 and 8. It is essential to note that Gradient, whose client selection method is identical to IMCS, is excluded from this comparison of selection effectiveness. The average quality of clients selected by IMCS is better than that of the other baselines, and it achieves superior regret performance. Taking the scenario with a high non-IID degree ($\alpha = 0.1$) as an example: On the MNIST dataset, at epoch 200, the regret of IMCS is 50.41% lower than Thompson, 86.99% lower than Importance, and 86.9% lower than Heuristic. On the CIFAR-10 dataset: At epoch 200, the regret of IMCS is 53.82% lower than that of Thompson, 83.38% lower than that of Importance, and 84.3% lower than that of Heuristic.

4) *Sensitivity Analysis:* We conduct a sensitivity analysis on parameters ν_1 , ν_2 (within the aggregator's cost function), and the tolerance parameter ϵ_N (used in Newton's method for solving the aggregator's optimal policy). This analysis is performed on the MNIST dataset under the conditions $\alpha = 0.1$ and $K = 10$.

Sensitivity to ν_1 and ν_2 values (fixed ratio ν_1/ν_2): We first examine the impact of varying the absolute values of ν_1 and ν_2 while keeping their ratio ν_1/ν_2 constant. Defining a parameter set $nu\ set = [\nu_1, \nu_2]$, Fig 8(a) compares: $nu\ set\ 1 = [10^2, 10^{-7}]$ (our chosen set), $nu\ set\ 2 = [10^3, 10^{-6}]$, $nu\ set\ 3 = [10^4, 10^{-5}]$. Results indicate that varying the absolute values of ν_1 and ν_2 has no significant impact on accuracy when their ratio ν_1/ν_2 remains fixed.

Sensitivity to the ν_1/ν_2 ratio: We then analyze the effect of changing the ν_1/ν_2 ratio itself. Fig. 8(b) shows results for: $nu\ set\ 4 = [10^2, 10^{-5}]$ ($\nu_1/\nu_2 = 10^7$), $nu\ set\ 5 = [10^2, 10^{-3}]$

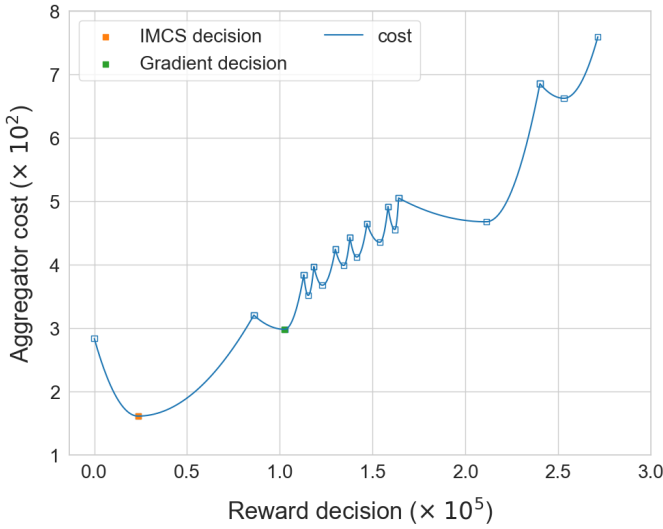


Fig. 9. Aggregator cost function when IMCS first reaches 90% accuracy on the MNIST dataset during training with $\alpha = 0.1$ and $K = 10$.

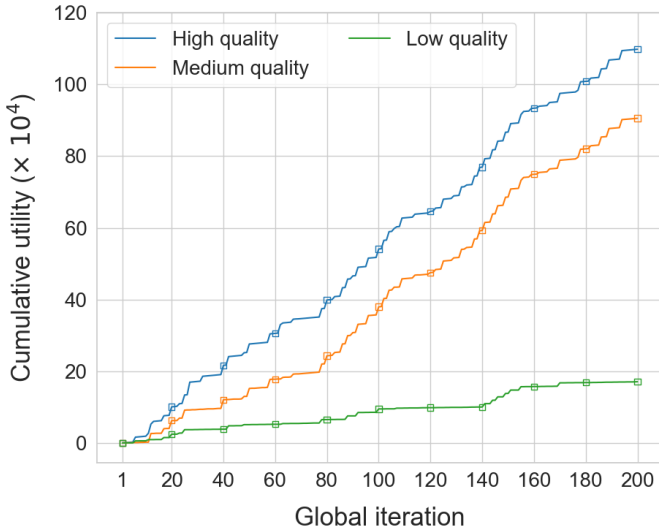


Fig. 10. Cumulative utility for clients with varying data quality on the MNIST dataset during training with $\alpha = 0.1$ and $K = 10$.

($\nu_1/\nu_2 = 10^5$), *nu set 6* = $[1, 10^{-3}]$ ($\nu_1/\nu_2 = 10^3$), *nu set 7* = $[1, 1]$ ($\nu_1/\nu_2 = 1$). The results demonstrate that lower ν_1/ν_2 ratios lead to lower final accuracy. Compared to *nu set 4* to *7*, our chosen *nu set 1* ($\nu_1/\nu_2 = 10^9$) achieves a maximum accuracy improvement of 2.19%. However, the impact of the ν_1/ν_2 ratio on accuracy becomes negligible once the ratio falls below approximately 10^7 .

Sensitivity to Newton's method tolerance ϵ_N : We evaluate the tolerance parameter ϵ_N for Newton's method by comparing accuracy at values: $\epsilon_N = 1, 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}$. As shown in Fig. 8(c), we find that reducing ϵ_N below 10^{-4} yields no significant further improvement in accuracy. Consequently, we select $\epsilon_N = 10^{-4}$.

Sensitivity to λ in Eqn. (8): Finally, we compared the global model accuracy for λ values of 2, 4, 6, 8, and 10. As shown in Fig. 8(d), the results indicate that the choice of λ has an

effect on both the final accuracy and convergence stability. Specifically, lower values such as $\lambda = 2$ and $\lambda = 4$ lead to slightly lower final accuracy. Conversely, values of $\lambda = 6, 8$, and 10 all achieve higher and more stable accuracy. We find that enhancing λ above 8 yields no significant further improvement in accuracy. Consequently, we select $\lambda = 8$.

5) Stackelberg Equilibrium Point Analysis: Fig. 9 illustrates the cost function of the aggregator when the global model achieves an accuracy rate of 90% for the first time. This function presents multiple local optimal solutions, which poses a challenge to simple optimization methods such as gradient descent. In contrast, our IMCS mechanism, which benefits from a theoretical analysis of the cost function's properties, can systematically evaluate all potential optimum points, thereby identifying the global minimum and ensuring the most cost-effective reward strategy. This verifies the robustness of IMCS.

Fig. 10 reveals the results of this optimal strategy from the client's perspective. We track the cumulative utility of high-quality, medium-quality, and low-quality clients (relative to the top 10 clients with data quality). The results clearly show that over time, IMCS successfully allocates significantly more rewards to higher-quality clients. This confirms that the followers respond to the leader's strategy as expected, and IMCS encourages the contributions that are most beneficial to the overall model. In conclusion, these two perspectives jointly provide a comprehensive verification of the SE point and its practical effectiveness.

VII. CONCLUSION

In this paper, we propose IMCS, an incentive mechanism for FL based on CMAB and Stackelberg game theory. IMCS comprises two key components: first, it frames client selection with unknown data quality as a CMAB problem, implementing a CUCB-based framework to balance exploration and exploitation, thereby maximizing global data quality. Second, it models the interaction between the aggregator and clients as a Stackelberg game, defining the utility functions of the aggregator and clients based on FL convergence analysis and data contributions. We theoretically prove the existence and uniqueness of the Stackelberg equilibrium and derive optimal strategies for all participants. Furthermore, we demonstrate a sublinear worst-case upper bound for our client selection framework compared to optimal selection. Finally, we conduct extensive experiments on the MNIST and CIFAR-10 datasets.

These experimental results validate the effectiveness and efficiency of IMCS. In particular, the consistent performance gains observed across both MNIST and CIFAR-10 datasets suggest the robustness and generalizability of IMCS. These findings demonstrate that IMCS offers a practical and efficient approach to incentivizing high-quality data contribution in FL environments, ultimately leading to improved model performance and reduced training costs. Future work will focus on extending IMCS to more complex and heterogeneous data distributions, exploring adaptive learning rate strategies within the CUCB framework, and investigating alternative game-theoretic models to further refine the incentive mechanism.

Furthermore, two significant extensions would be to adapt our framework to handle: (1) explicit client churn, where the client set $\mathcal{W}$ is time-varying, and (2) non-stationary environments caused by client mobility, where a client's expected quality λ_i^t drifts over time. Both challenges would require moving beyond the standard CMAB framework, presenting non-trivial future research directions. Building on this foundation, we will also extend our evaluation to larger-scale deployments with hundreds or thousands of clients and consider more dynamic scenarios, such as those involving heterogeneous devices and time-varying client quality. We also plan to analyze the impact of varying client populations and network conditions on the performance of IMCS in real-world FL deployments.

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Jianxiong Guo (Member, IEEE) received his Ph.D. degree from the Department of Computer Science, The University of Texas at Dallas, in 2021, and his B.E. degree from the School of Chemistry and Chemical Engineering, South China University of Technology, in 2015. He is currently an associate professor at the Advanced Institute of Natural Sciences, Beijing Normal University, China. He is a member of IEEE/ACM/CCF. He has published more than 100 peer-reviewed papers and has been a reviewer for many famous international journals/conferences. His research interests include edge intelligence, IoT, distributed machine learning, and (Online) Optimization.



interests include network artificial intelligence, distributed machine learning, and online optimization.

Juncheng Wang (Member, IEEE) received the B.Eng. degree in Electrical Engineering from Shanghai Jiao Tong University, Shanghai, China, in 2014, the M.Sc. degree in Electrical and Computer Engineering from the University of Alberta, Edmonton, AB, Canada, in 2017, and the Ph.D. degree in Electrical and Computer Engineering from the University of Toronto, Toronto, ON, Canada, in 2023. He is currently an Assistant Professor with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. His research



Xingjian Ding received the B.E. degree in electronic information engineering from Sichuan University, Chengdu, China, in 2012, the M.S. degree in software engineering from Beijing Forestry University, Beijing, China, in 2017, and the Ph.D. degree from the School of Information, Renmin University of China, Beijing, in 2021. He is currently an associate professor at the College of Computer Science, Beijing University of Technology, Beijing. His research interests include Approximation Algorithms Design, Edge Computing, and Federated Learning.



Deying Li is a professor at the Renmin University of China. She received the B.S. degree and M.S. degree in Mathematics from Huazhong Normal University, China, in 1985 and 1988 respectively. She obtained the PhD degree in Computer Science from the City University of Hong Kong in 2004. Her research interests include wireless networks, ad hoc & sensor networks, mobile computing, distributed network system, Social Networks, and Algorithm Design, etc.



Jiajun Wang received the B.S. degree from Shaanxi University of Science and Technology, in 2023, and is currently pursuing the M.E. degree with the College of Computer Science, Beijing University of Technology, Beijing, China. His research interests include federated learning and edge computing.



Weili Wu (Fellow, IEEE) received the Ph.D. and M.S. degrees from the Department of Computer Science, University of Minnesota, Minneapolis, MN, USA, in 2002 and 1998, respectively. She is currently a Full Professor with the Department of Computer Science, The University of Texas at Dallas, Richardson, TX, USA. Her research mainly deals with the general research area of data communication and data management. Her research focuses on the design and analysis of algorithms for optimization problems that occur in wireless networking environments and various database systems.